



Smart Hydroponic System

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1. Abstract

The Smart Hydroponic System is an automated soil-less farming system designed to improve plant growth using sensors and decision-making software. Traditional farming comes with challenges like limited land, unpredictable weather or the lack of water resources. The Smart Hydroponic System solves these challenges by growing the plants in water that is rich with nutrients without any soil needed. The system monitors the environmental factors that are important for plant growth such as water levels, pH, electroconductivity, nutrients and light. After collecting this data, the Smart Hydroponic System adjusts some to create the best conditions for plant growing.

Controller checks the sensor data and based on that data hardware adjustments through pumps or lights are made. The need for manual work is reduced when compared to traditional farming. An easy-to-use interface allows the users to check the system in real-time so they are aware of the state of the system and are able to make smarter decisions. The Smart Hydroponic System promotes sustainability as it allows to reduce water and energy use. The Smart Hydroponic System is a solution for urban farming and other environments like greenhouses, that can also grow with. Thus, the system improves farming efficiency and also helps with food supply and saving resources.

2. Problem Definition and Background

Problem Statement

There are many problems in traditional farming such as the limited land, changing weather and water waste. Because of such issues, there is a problem of uneven crop growth and reduced harvests. Hydroponic farming tries to solve challenges by allowing plants to grow not in soil, but a nutrient-rich water solution. However, traditional hydroponic farming requires manual monitoring and adjustments of the important things like pH or water levels. Management by hand of such a system can require a lot of work, and may cause mistakes. The Smart Hydroponic System tries to solve this problem by automating the control of the system through using sensors and AI-driven decision-making using the information from the sensors

Background

Hydroponic farming has become popular due to its ability to grow plants in controlled spaces, especially in areas where traditional farming is not efficient due to environmental problems. However, managing these systems by hand is challenging, and requires constant monitoring of the important factors to ensure it works well for plant growth. Mistakes or slow adjustments in nutrients or water levels will result in poor plant growth.

The **Smart Hydroponic System** uses automation to solve these challenges. Sensors collect real-time data on the growing environment, which is handled by AI to make right adjustments, ensuring plants receive the right amount of water, nutrients, and light at all times. By doing this, the Smart Hydroponic System reduces human error, saves time, and works more efficiently. This makes adopting hydroponic farming easier and more available.

Relevance

It is no secret that the world is facing many challenges around food production and agriculture. The Smart Hydroponic System offers an effective and sustainable solution to these problems, allowing, for example, to grow more food in urban areas where space is limited. By automating the process, it saves water and energy usage while improving crop growth. As climate change continues to harm traditional farming, this system provides a more reliable and efficient way for growing food in controlled environments, which are, most importantly, not heavily affected by climate change.

3. Project Objectives and Goals

Objectives

Build an Automated Hydroponic System: The main goal of this project is to create a smart hydroponic system that automatically checks and controls the environmental factors that are important for plant growth such as the water levels, nutrient mix, pH balance, light intensity, and humidity.

Integrate AI and Sensors for Real-Time Monitoring: The system will use different sensors to gather a variety of real-time data regarding the environment. AI algorithms will study this data, and then make adjustments to improve the conditions for plant growth. This will ensure steady and better growing conditions for plants, with minimum human work required.

Create an Easy-to-Use Interface: One of the key objectives is to create an easy-to-use interface that allows users to control the system remotely. The interface allows users to view the work of the system, receive notifications about the system status and control the system remotely.

Promote Sustainable Farming: One of the other key objectives is to minimize the usage of important resources such as water and energy by automating watering and nutrient delivery.

Goals

Maximize Crop Yield and Consistency: Because the management of the system will be done by an AI system, it will be precise and will lead to increased crop yields and the consistency of the crop, which are one of the goals of this system.

Minimize Human Labor and Error: Because the system is highly automated, minimal human work is required. This reduces mistakes and reduces the work required to manage the hydroponic farm.

Enhance Resource Efficiency: Because of the accuracy of the system, the amount of water and nutrients used will be almost only as much as needed. This will contribute to the sustainability of the system and its positive effect on the environment.

Scalability and Flexibility: The system will be designed to be scalable, allowing it to be adapted for different farming sizes, from small-scale urban gardens to larger commercial farms. It should also be flexible enough to not cause any problems with future features, such as new sensors or control mechanisms.

Provide Data-Driven Insights: The system will offer detailed data on plant growth and system performance, helping users make better decisions and improve future harvests.

4. Stakeholder Analysis

Primary Stakeholders

Farmers and Urban Growers

Farmers and urban growers are interested in increasing crop yields while minimizing manual labor and resource consumption. This will lead to more profits. They need a reliable, easy-to-use system that automates the management of key environmental factors such as water or pH levels. They are also concerned with reducing operating costs and improving the sustainability aspect of their farming.

Regulatory Bodies

Regulatory bodies are focused on ensuring the Smart Hydroponic System is up to all the food safety, environmental, and agricultural regulations. This includes ensuring that the system does not have any negative impact on the local ecosystem.

Agricultural Equipment Manufacturers

Manufacturers are interested in producing high-quality and durable equipment that works well with the Smart Hydroponic System. They can establish long-term relationships with farmers and the system developer to provide the needed products.

Secondary Stakeholders

Consumers

Consumers need a high-quality and sustainable product. They are interested in knowing that the food they buy is grown in an environmentally friendly way, and without using harmful chemicals. Consumers are also getting more and more interested in supporting local, urban farming initiatives that reduce food waste, are usually more sustainable and give higher-quality products.

Investors and Financial Backers

The main concern of the investors is the financial potential and the scalability of the system. They are mainly concerned if the system is financially profitable and if it has the potential to expand to larger markets..

Research and Development Teams

These stakeholders are focused on the system's innovative potential and performance. They are interested in developing new features and improving the precision of all the parts of the Smart Hydroponic System. Their goal is to make sure that the system remains innovative and offers more and more useful functions over time.

5. System Requirements

Functional Requirements

The hydroponic system must follow several functional requirements to ensure some proper and optimized plant growth.

1. Water Management

The system must automatically measure and maintain water levels.

The system has to keep the water pump running continuously to circulate water, and deliver water to the plants.

2. Nutrient Level Monitoring and Adjustment

The system must monitor the nutrient levels in the water with some sensors like pH and EC.

The system shall activate pH and EC balancing valves to keep the nutrient levels within the optimal range for plant growth.

3. Lighting Control

The system must deliver initial lighting when the system starts.

The system has to automatically adjust light intensity based on the feedback from the light sensor.

4. Temperature and Humidity Control

The system has to measure the temperature and humidity within the system.

5. User Interface and Notifications

The system has to include a user-friendly interface for controlling and receiving information about the plants in the system.

Non-Functional Requirements

These requirements define the quality and operational standards for the hydroponic system:

1. Performance

The system must process sensor inputs and adjust settings in real time, within lowest response time.

The water and nutrient pumps have to be capable of delivering consistent flow rates under varying conditions.

2. Reliability

The system must handle simultaneous commands without performance degradation.

3. Maintainability

The system must allow for easy replacement of components such as pumps, sensors, and relays.

Software updates have to be easy to apply without requiring more than 30 minutes.

4. Environmental Considerations

The system has to be designed to operate efficiently with minimal energy consumption.

Components must be water-resistant or water-proof to make sure they will operate in wet conditions.

5. Usability

The user interface should be easily understandable, readable and take not a lot of time to learn how to use.

The system settings must be able to be easily changed by using the user interface and the documentation.

6. Safety

Electrical components must be isolated with a hydrophobic isolator to prevent accidents due to water exposure.

6. System Architecture

The Smart Hydroponic System consists of four main components: circulation subsystem, controlling unit, sensors unit, and reservoir. Each component has its own importance in keeping the system running and working properly. These components work together to satisfy the essential conditions for the optimized plant growth.

Below you can see the key components of the system ([See figure 6.1](#)). As well as the system architecture in the form of a circuit diagram ([See figure 6.2](#)).

1. Circulation Subsystem

This Subsystem makes sure that there is a continuous supply of water and nutrients to the plants. This subsystem ensures that the nutrients are evenly distributed to the plant. It consists of a water pump, valves and nutrient solution injectors.

2. Controlling Unit

Just like the human body a system needs a brain. The controlling unit acts like the brain of the system. It is constantly waiting for data from the sensors, later processes that data, and after processing it gives feedback to the actuators. For example if the pH levels in the water have dropped, the controlling units recognize that and send a signal to the valves to pour some more pH controlling liquid into the water by balancing it.

3. Sensors Unit

If we once again bring an example of the human body, sensors will be our eyes, ears etc. Sensors are constantly gathering data about the conditions in the system and sending it to the controlling unit, enabling it to understand what is going on in the system.

4. Reservoir

The Reservoir acts like a big storage for storing all the necessary liquids, balancers, fertilizers for the system.

Interaction Between Subsystems

All the subsystems listed above would be not useful if they worked without communication between each other. The Sensors send data to the controlling unit with continuous updates about the conditions in the system. The Controlling Unit after receiving that information decides what is necessary for balancing the conditions, and sends a feedback to the circulation unit to complete a certain action, which brings the conditions to the optimised state ([See figure 6.3](#)).

Block Definition Diagram

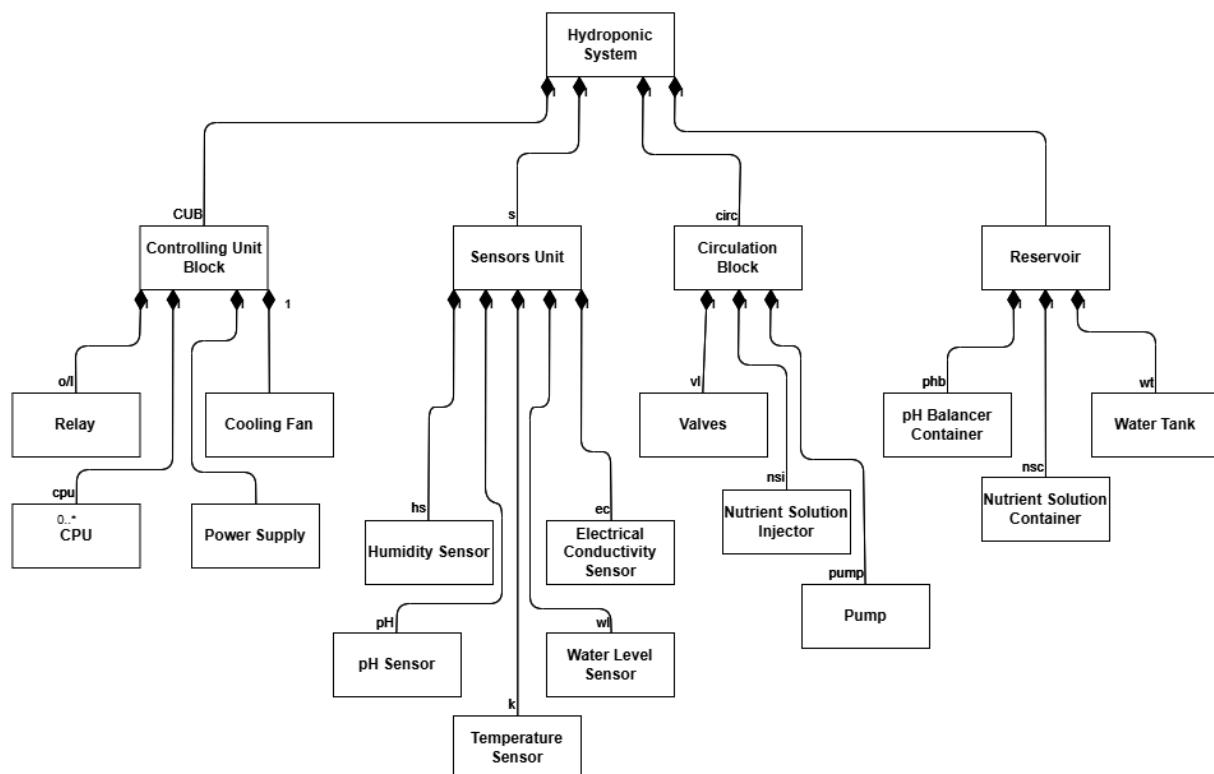


Figure 6.1 (Ashot Nahapetyan)

The Block Definition Diagram shows the components of our Smart Hydroponic System. The Sensors Unit constantly sends data about conditions in the system. A reservoir system holds water and nutrients. The Circulation Unit makes sure the water and nutrients reach the plant. The controlling unit makes all those elements work together and ensures communication between them.

Circuit Diagram

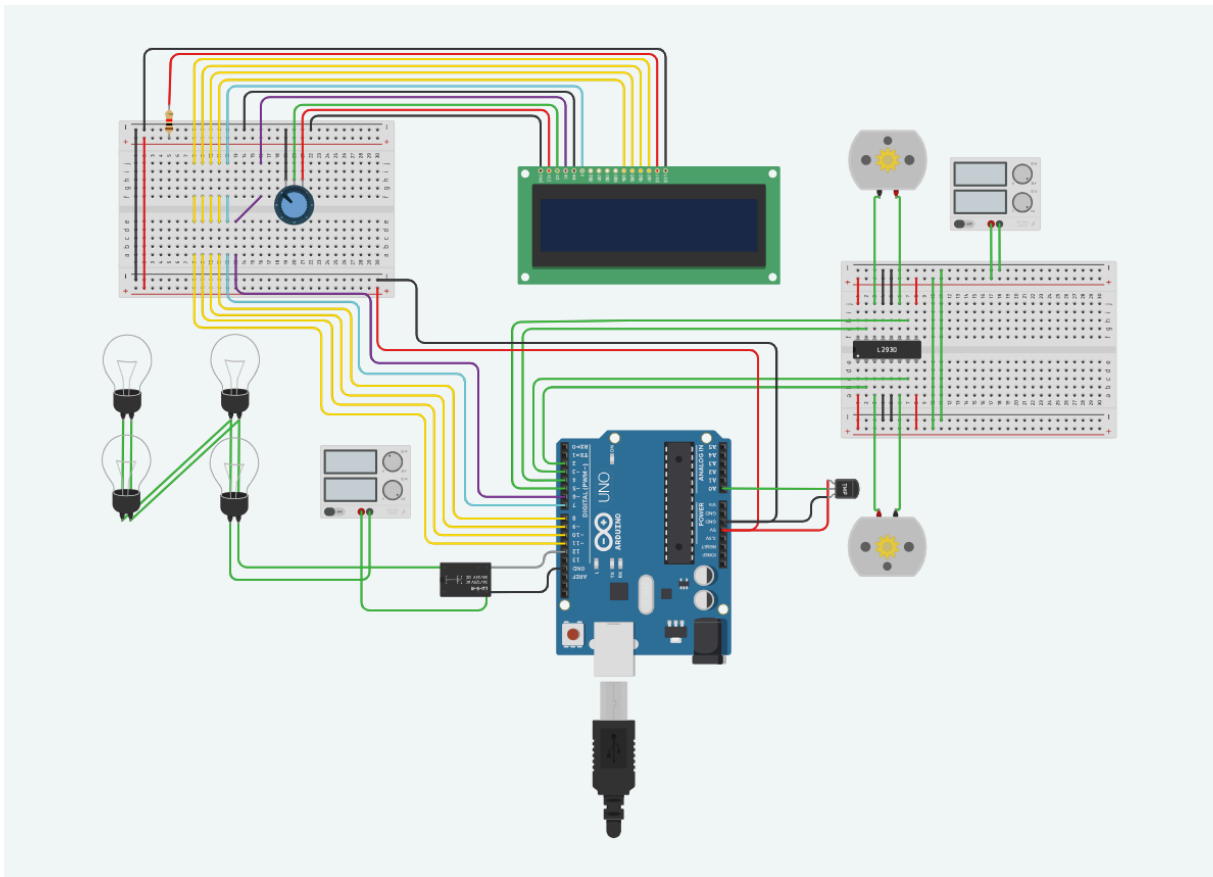


Figure 6.2 (Ashot Nahapetyan)

The circuit diagram shows the architecture of hardware in our hydroponics system, illustrating the exact connections with wires from sensors to microcontroller, and from controller to actuators, and overall look of the hardware assembled.

Functional Diagram

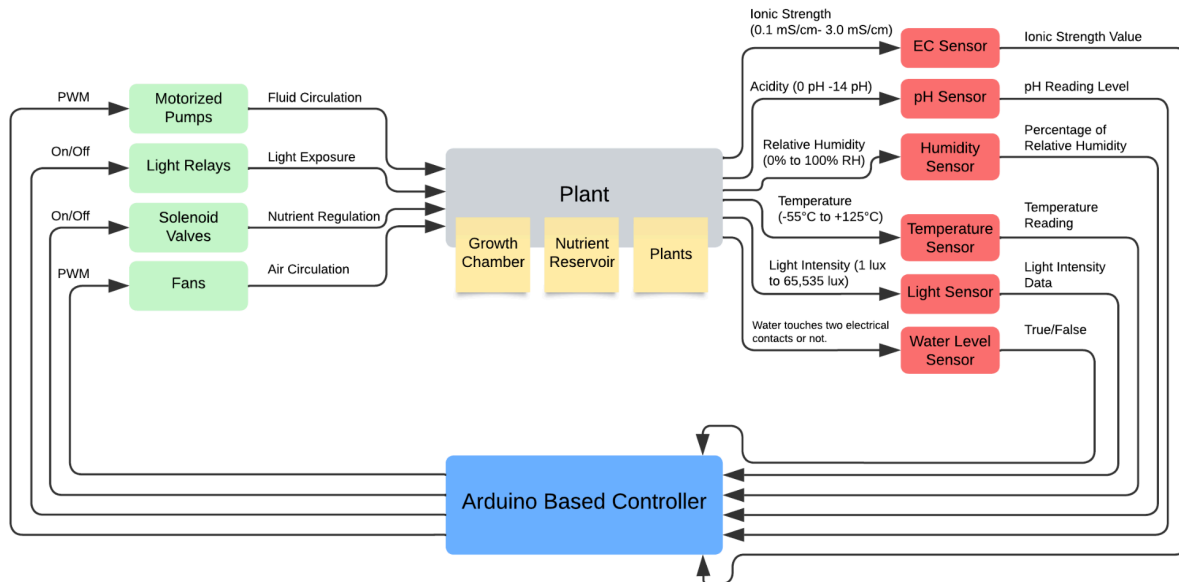


Figure 6.3 (Aghasi Grigoryan, Lyusi Saiyan, Ashot Nahapetyan)

Sensors:

Sensor → Controller → Actuator

The Sensors monitor the plant's environment, detecting changes in key parameters like light, humidity, temperature, pH, etc. These sensors send continuously collected data to the controller, which adjusts actuators to optimize conditions.

Types and Input Signals:

pH sensor measures the acidity of the water. **Chemical sensor | Acidity Analog Signal.**

Light Sensor Detects the intensity of light in the environment. **Optical sensor | Light Intensity Analog Signal.**

Humidity Sensor Monitors air humidity levels. **Environmental sensor | Humidity Analog Signal.**

Temperature Sensor Measures air temperature. **Environmental sensor | Temperature Analog Signal.**

Electrical Conductivity sensor Measures ionic strength. **Chemical sensor | Ionic Strengths Analog Signal.**

Water Level: Checks water level. **Fluid sensor**

Actuators:

Controller → Actuator → Plant

The Controller sends electrical control signals to actuators, telling them to manage devices such as UV lights, fans, water pumps, and nutrient pumps. These actuators convert the signals into actions that regulate key processes like light exposure, climate control, water circulation, and nutrient delivery.

Types and Input Signals:

Light Relays: Turn on or off UV lights via a relay. **Electric actuator | On/Off Signal.**

Motorized Pumps: Activates or deactivates, adjusts the intake of the pump ensuring the proper fluid circulation. **Hydraulic actuator | Pulse Width Modulation.**

Solenoid Valves: Turns the nutrient pump on or off. **Hydraulic actuator | On/Off Signal**

Fans: Adjusts air flow rate by controlling the fan speed. **Electric actuator | Pulse Width Modulation.**

Plant:

Actuators → Plant → Sensors

The **plant** is the environment where plants grow, relying on the system's regulated water, nutrients, light, and environmental conditions. Sensors and actuators work together to optimize growth.

Controller:

Sensors → Controller → Actuator

Controller gets the information from the sensors about the humidity levels, pH levels, temperature, UV light levels from the sensor.

For example, if the humidity level is not enough, the controller sends a signal to the circulation unit to open the valves and increase the humidity level. This continues until the reading from the sensor suggests that the humidity level is good. The same way, if the pH levels are low, the controller sends a signal to the actuator to pour some pH balances and increase the pH levels until sufficient. If the light levels are low, the controller sends a signal to the UV lights generator to increase the light until it's sufficient. Also, if the humidity or the temperature is too high, the controller sends a signal to the actuator (the ventilator) to turn on. The controller receives analog signals from the sensors (the humidity, pH, temperature, and light levels), and processes them by dividing the original values to some constants to get a range of values easier for usage. This ensures that the controller works with standardized and manageable data.

7. Requirements Validation and Risk Analysis

Requirements Validation

1. Component Testing

Sensors: Each sensor (water level, pH, light, EC) will be individually tested to make sure there is a precise data collection. For example, the pH sensor will be tested by exposing it to different acidity levels to verify its accuracy.

Control Systems: Pumps, nutrient valves, and lighting will be tested to make sure they respond correctly to the commands sent by the system. For example, turning on/off water pumps when required and adjusting light intensity appropriately.

2. Integration Testing

We will start with assembling the whole system with components, making sure they nicely fit together, there are no gaps, and the system components nicely communicate together. We can achieve this by running test commands to see if sensors trigger correct actuators. If they do, that means the connection between sensors, controllers and actuators is nicely done.

3. User Acceptance Testing

The system will be tested by users in real world scenarios, and professionals will test whether the system is easy to use and all the requirements are met. The feedback from these tests will improve the further development.

Risk Analysis

1. Technical Risks

Sometimes sensor data can be incorrect because sensors often malfunction. As they provide inaccurate information it will disrupt the systems operation. For example, if the sensor sends false information about the fertilizer levels in the water, and the controller adds more fertilizer to the system, that can easily burn the roots of the plants.

2. Operational Risks

The electricity we use is not always reliable. For example, some power outages can occur and cause the whole system to stop working. Those system interruptions can cause a lot of damage to the plant's health. We can deal with this by adding a backup power supply that will ensure that the system will work during power outages.

3. Maintenance Risks

Over time some components of the system will wear out which will lead to system malfunction. For example, a sensor can wear down and start measuring a level of some condition incorrectly. That will send some misinformation to the controller, which will trigger the wrong actuator and cause harm to the plant or even to the system itself.

8. Life Cycle Considerations

The life cycle of the Smart Hydroponic System includes several stages like concept, development, production, utilization and retirement.

1. Concept Phase

In this phase we focus on defining our system's purpose. During this phase we have to do a nice research and understand what our system needs and what are the requirements. One has to take into consideration the preferences of stakeholders. Also several studies about plants, costs and risks should be conducted.

2. Development Phase

During this phase, the first steps for building the system are done. It all starts with a sketch of the system, prototypes should be built and each subsystem should be tested independently to make sure they will work properly in the future phases.

3. Production Phase

Whenever we are done with developing the system and are sure that the system won't break anymore we can start the production phase. In this phase the components are assembled, tested, and prepared for installation. Installation and operation guides are finalized and the system is ready to be deployed.

4. Utilization Phase

In the utilization phase the system is deployed and used. The sensors are being calibrated for the customers preferences, and different adjustments are made to ensure a smooth operation. Also system updates will be done over time to make sure the system doesn't age out, because some new studies about plants are being published daily.

5. Retirement Phase

At the end of its life, the system has to be properly retired. All the components must be properly recycled to minimize the environmental impact, because the chemicals in the system might be harmful for the environment. Also some parts of the system are reusable and they can be properly disassembled and used for new systems.

9. Project Timeline and Management Plan

The development of the project will be divided into five phases. These phases include planning, design, development, testing, development and maintenance.

Project Phases and Timeline

1. Phase 1: Requirements Gathering and System Design (2 weeks)

In this phase stakeholders data will be gathered to clarify requirements. Then the system design and architecture will be created, and high-level diagrams and sketches to outline components and visualize the next stages.

2. Phase 2: System Development (4 weeks)

With the initial requirements system components can be developed. Each component will undergo individual testing to ensure functionality making sure that the components will be ready for integration.

3. Phase 3: Integration and System Testing (3 weeks)

The developed components will be integrated and tested to ensure seamless operation. The system's performance will be tested under various conditions, and every problem that arises during testing will be fixed.

4. Phase 4: User Testing and Feedback (2 weeks)

Users will test the system to assess its usability. Feedback will be collected to improve system performance. By the end of this phase, a finalized user guide will be completed, addressing any constraints identified during testing.

5. Phase 5: Final Deployment and Maintenance (2 weeks)

With the system ready it will be deployed including its calibration to user preferences, and further maintenance. During this phase some unexpected challenges might occur. These will help to improve the system to excellence.

Management Plan

1. Resource Management - The project will be managed by a team of engineers, testers and specialists.
2. Scheduling - Use of different time management applications can be used to keep track of time. For example, to do lists, Gantt Charts etc.
3. Stakeholder Communication - Stakeholders will be provided with updates about the project on a weekly basis.

10. Implementation and Testing Plan

Implementation Plan

The implementation of the system will follow a structured plan focusing on the integration processes and strategies that verify that the system meets the requirements.

1. Component Design and Development

Each component of the system will be developed individually, tested for functionality, and later optimized for future integration into the system.

2. Subsystem Integration

After developing the components, subsystems will be developed and integrated. Their interaction will be tested to ensure they work together effectively.

3. System Calibration

This step is essential to adjust the system to satisfy user preferences. Proper calibration minimizes error margins, reducing resource consumption and energy use ensuring tasks are completed efficiently.

4. Deployment

Finally the system will be deployed, the user will be trained to operate it, and support will be provided to address any performance issues.

Testing Plan

To ensure that the system functions as expected, a comprehensive testing plan will be implemented such as integration testing, user testing and user acceptance testing.

Unit Testing

Objective: Verify the functionality of individual components (sensors, pumps, AI algorithms) before integrating them into the system.

Tests:

Sensor Testing: Ensure that each sensor (pH, water level, light) accurately measures environmental conditions and transmits data to the control unit.

Pump and Actuator Testing: Test water pumps, lighting, and nutrient dispensers to confirm they operate correctly when triggered by the system.

AI Algorithm Testing: Simulate different environmental conditions to ensure the AI makes appropriate decisions for adjusting water, nutrients, or light.

Team: Hardware and software teams will run unit tests for their respective components.

Timeline: 2 weeks.

1. Integration Testing

Objective: Ensure that all hardware and software components work properly together as a whole system.

Tests:

Sensor-Software Integration: Make sure the data measured by the sensors is correctly sent to the controller, without any information loss.

Software-Hardware Integration: Verify that the control system accurately triggers actuators based on decisions.

End-to-End Testing: Run the system in a test environment to simulate different scenarios and ensure the system reacts appropriately by adjusting the necessary settings.

Team: Cross-functional teams including hardware and software engineers will collaborate to ensure seamless integration.

Timeline: 3 weeks.

2. User Acceptance Testing (UAT)

Objective: Ensure that the system meets the needs of the end users and functions as expected in a real-world environment.

Tests:

Usability Testing: Users such as farmers or operators will test the system and provide feedback on its user-friendliness, functionality of the user interface and how adaptable the system is in a changing environment.

Performance Testing: The system's ability to maintain optimal growing conditions will be tested under real farming circumstances. Adjustments to water, nutrients, and light based on sensor data will be monitored.

Feedback Loop: User feedback (issues or suggestions) will help to make final adjustments.

Team: End-users, UI/UX designers, and engineers will collaborate to clarify the system based on feedback.

Timeline: 2 weeks.

3. Performance Testing

Objective: Evaluate the system's ability to maintain performance standards under different conditions such as, heavy usage or fluctuating environmental factors.

Tests:

Stress Testing: Simulate high-demand scenarios, such as rapidly changing environmental conditions, to make sure that the system can handle heavy data loads and make real-time adjustments.

Reliability Testing: Run the system for long hours to check that it maintains a 99.9% uptime and accurately adjusts environmental factors consistently.

Team: Software engineers and System administrators will track performance results.

Timeline: Ongoing after initial deployment.

4. Security Testing

Objective: Ensure that the system is secure by protecting all data transmissions between the sensors, control unit, and user interface from unauthorized access.

Tests:

Data Encryption Testing: Confirm that all data transmitted between the sensors and the user interface is provided with some level of encryption.

User Authentication Testing: Ensure that only authorized users can access the system and modify settings.

Team: Security specialists and software developers will conduct security testing.

Timeline: 1 week.

Sequence Diagram

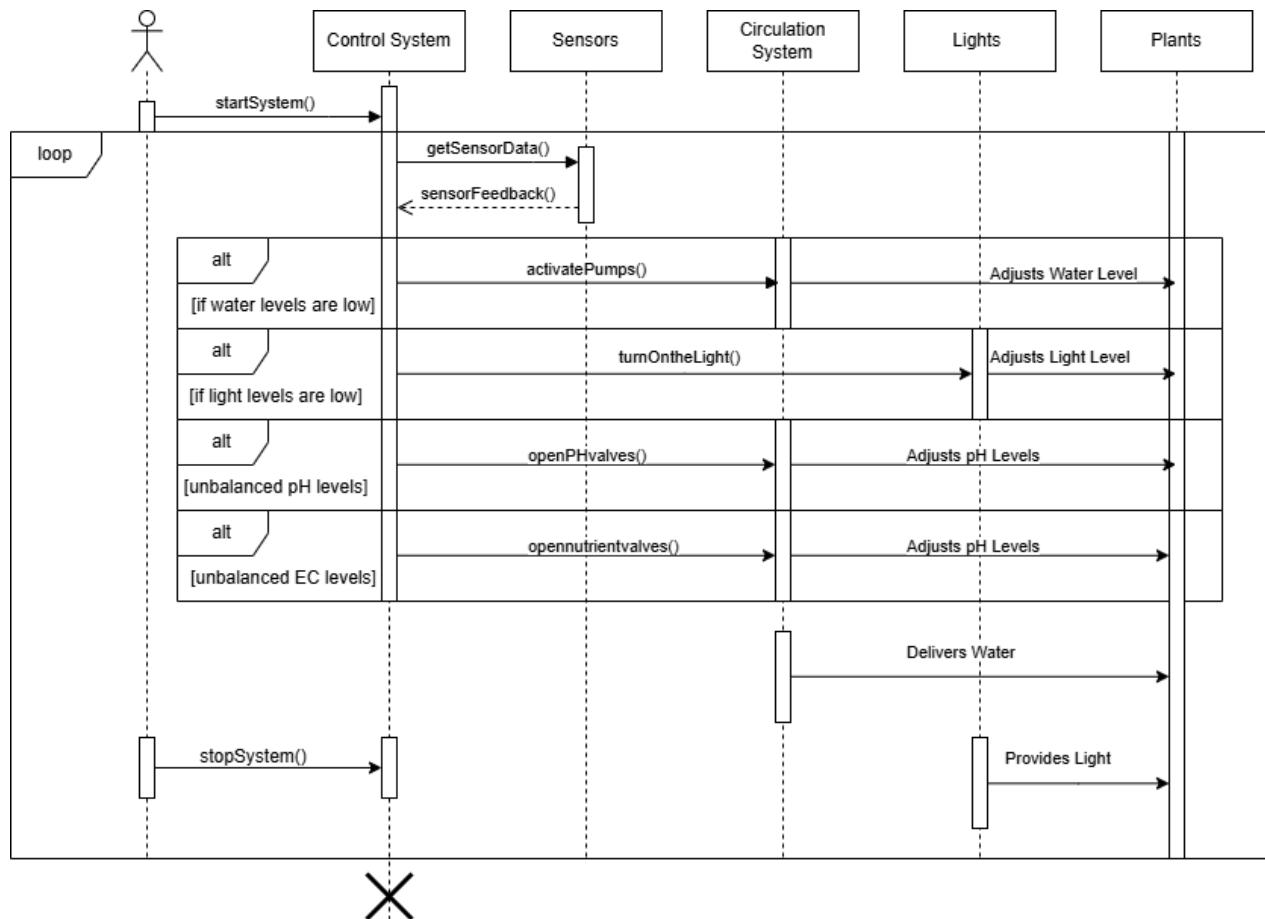


Figure 10.1 (Aghasi Grigoryan)

This sequence diagram shows the main interaction flow in smart hydroponic systems. The system is turned on when the user turns it on. The system constantly monitors important metrics for the plants to grow in a hydroponic system such as the pH levels, water levels and even the light level which is regulated through UV lamps. If the plants need more water, the sensors capture that and send a signal to the pump to add more water. The same principle works with light levels and pH levels.

State Machine Diagram

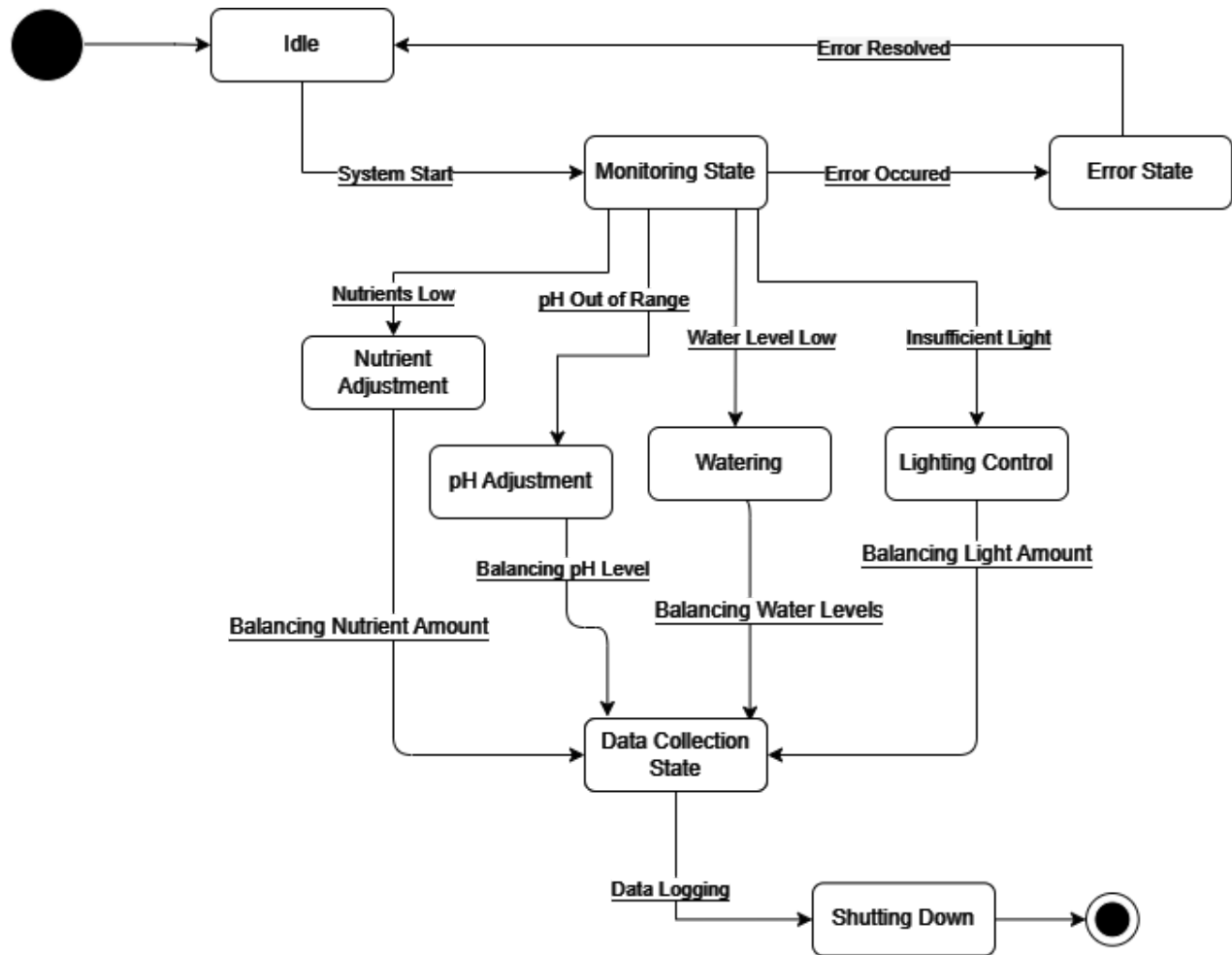


Figure 10.2 (Lyusi Saiyan)

The state diagram represents the workflow of the smart hydroponic systems. It begins in an idle state. After being activated, the system moves to the monitoring state. In the monitoring state, it constantly checks the main parameters such as the pH level, water level, and the lighting level. If any of these need adjustment, the system makes the adjustments and then goes back to the data collection state. If an error is detected, the system moves to an error state until the error is resolved. The system works until shut down, where it goes to the shutting down state.

11. Integration and Deployment Plan

Integration Plan

1. Hardware and Software Integration:

Objective: Ensure seamless integration between sensors, control units and actuators (pumps, lights) and software/AI system.

Steps:

Connect sensors to the control unit for data collection.

Link actuators to the control system to respond to AI commands.

Test data flow ensuring accurate sensor readings and correct actuator responses.

Outcome: A fully integrated system where hardware and software components interact smoothly.

2. User Interface Integration:

Objective: Provide users with real-time data and control options.

Steps:

Integrate real-time sensor data into the user interface.

Enable control through the UI for manual adjustments if needed.

Outcome: A user-friendly interface that allows monitoring and control of the system.

3. System Testing:

Objective: Test the full system under real-world conditions.

Steps:

Simulate different scenarios (low water, improper pH) to test the system's response.

Verify that the system adjusts conditions correctly based on sensor input.

Outcome: A fully functional and tested system ready for deployment.

Deployment Plan

1. System Setup:

Objective: Install the system in the operational environment.

Steps:

Install sensors, pumps, and lights, and configure the control unit.

Perform initial calibration of sensors and test the system in the field.

Outcome: A fully operational system installed in its target environment.

2. User Training:

Objective: Teach users to operate and monitor the system.

Steps:

Conduct training sessions on using the interface and system controls.

Provide manuals and demonstrate real-time system adjustments.

Outcome: Trained users who can independently manage the system.

3. Post-Deployment Support:

Objective: Ensure smooth operation and address issues after deployment.

Steps:

Offer ongoing technical support and scheduled maintenance.

Monitor system performance and provide software updates as needed.

Outcome: Continued system reliability and user satisfaction through ongoing support.

12. Conclusion

The Smart Hydroponic System offers a practical and innovative solution to modern agricultural challenges. By automating the control of water, nutrients, and light through advanced sensors and AI algorithms, it improves plant growth efficiency and reduces resource use minimizing manual labor and optimizing environmental conditions, therefore making it ideal for urban and large-scale farming.

With a thorough testing and validation process, the system ensures reliability and adaptability for various farming applications. It's flexible and versatile design allows it to evolve with future technological advancements, offering a sustainable and efficient approach to food production.

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